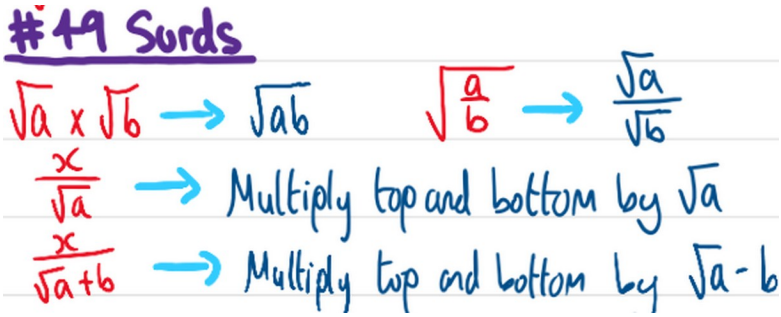
Worksheet 22 · Surds IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Simplify surds • Expand brackets containing surds • Rationalise the denominator of a fraction		
Grade: / 53	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes



Questions *Surds*

Question 1 (4 marks)

(a) Show that $\sqrt{45} + \sqrt{20} = 5\sqrt{5}$
Show your working clearly.

(2)

(b) Express $\frac{2}{\sqrt{3}-1}$ in the form $p + \sqrt{q}$
where p and q are integers.
Show your working clearly.

(2)

Question 2 (3 marks)

(a) Show that $\left(6 + 2\sqrt{12}\right)^2 = 12\left(7 + 4\sqrt{3}\right)$

Show each stage of your working.

(3)

Question 3 (3 marks)

(a) Rationalise the denominator of $\frac{a + \sqrt{4b}}{a - \sqrt{4b}}$ where a is an integer and b is a prime number.

Simplify your answer.

(3)

Question 4 (3 marks)

Given that $(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$ where x is a prime number and y is an integer,

find the value of x and the value of y.
Show each stage of your working clearly.

$x =$ .....

$y =$ .....

Question 5 (3 marks)

Express $\sqrt{48} + \sqrt{108}$ in the form $k\sqrt{6}$ where k is a surd.

.....

Question 6 (3 marks)

$a = \sqrt{8} + 4$

$b = \sqrt{8} - 4$

$(a - b)(a + b)$ can be written in the form $y\sqrt{4y}$

Find the value of y
Show your working clearly.

$y =$ .....

Question 7 (3 marks)

$(3 + \sqrt{a})(4 + \sqrt{a}) = 17 + k\sqrt{a}$ where a and k are positive integers.

Find the value of a and the value of k.

$a =$ .....

$k =$ .....

(1)

Question 8 (8 marks)

$\frac{1}{5^3} = 5^p$ $\quad 1 = 5^q$ $\quad \sqrt{5^3} = 5^r$

(a) Write down the value of

(i) p

$p =$ .....

(ii) q

$q =$ .....

(iii) r

$r =$ .....

(3)

(b) Show that $\frac{14}{\sqrt{245}} = \frac{2\sqrt{5}}{5}$

You must write down each stage of your working.

(2)

$\left(e - 2\sqrt{3}\right)^2 = f - 20\sqrt{3}$ where e and f are integers.

(c) Find the value of e and the value of f

$e =$ .....

$f =$ .....

(3)

Question 9 (3 marks)

(b) Rationalise the denominator of $\frac{4}{7 - \sqrt{5}}$

Show each stage of your working.

Give your answer in the form $a + b\sqrt{5}$ where a and b are fractions in their simplest forms.

(3)

Question 10 (4 marks)

Given that $8\sqrt{m} + \sqrt{49m} - \sqrt{9m} = k\sqrt{m}$
where k is an integer and m is a prime number,

(a) work out the value of k

$k =$ .....

(1)

(b) Show that $\frac{5 - \sqrt{18}}{1 - \sqrt{2}}$ can be written in the form $a + b\sqrt{2}$
where a and b are integers.
Show each stage of your working clearly.

(3)

Question 11 (3 marks)

Express $\frac{8}{\sqrt{5} - 1}$ in the form $\sqrt{a} + b$ where a and b are integers.
Show each stage of your working clearly.

Question 12 (4 marks)

Show that $\frac{\sqrt{12} - 1}{2 - \sqrt{3}}$ can be written as $4 + 3\sqrt{3}$

Show your working clearly.

Question 13 (3 marks)

Without using a calculator, show that $\frac{12}{\sqrt{2} - 1} - \left(\sqrt{2}\right)^5 = 2\sqrt{32} + 12$

Show your working clearly.

Question 14 (3 marks)

Show that $\frac{\sqrt{20} + \sqrt{80}}{\sqrt{3}}$ can be expressed in the form $\sqrt{a}$ where a is an integer.

Show your working clearly.

Question 15 **(3 marks)**

Show that $\frac{4+\sqrt{8}}{\sqrt{2}-1}$ can be written in the form $a+b\sqrt{2}$, where a and b are integers.

Show each stage of your working clearly and give the value of a and the value of b.